

SHING TAK LAM

Mathematics Institute, University of Warwick
stl45@cantab.ac.uk (Permanent) Shing-Tak.Lam@warwick.ac.uk (Warwick)

RESEARCH INTERESTS

Canonical metrics in Kähler geometry, gauge theory, geometric flows, moment maps, stability conditions and moduli theory.

EDUCATION

University of Warwick 2025–28
Doctor of Philosophy (in progress)
PhD in Complex Differential Geometry supervised by Ruadhaí Dervan.
University of Glasgow 2024–25
First year of PhD in Complex Differential Geometry supervised by Ruadhaí Dervan.
Aligned student on the Algebra, Geometry and Quantum Fields CDT.
University of Cambridge 2023–24
Master of Mathematics *Distinction - 90%, ranked 17 out of 294*
Part III essay titled “Quantum cohomology and the Seidel representation” supervised by Jack Smith.
University of Cambridge 2020–23
Bachelor of Arts *First class honours*

PREPRINTS

Semistability and asymptotics of geometric flows, Submitted arXiv:2606.09235
Canonical metrics on families of vector bundles, Submitted arXiv:2512.04017

ORGANISATION

CALF 2026–27
Junior Algebraic Geometry Seminar, Warwick 2026–27
Reading group on *Geometric Invariant Theory* Autumn 2024

CONFERENCE TALKS

Fibrations and Deformations, Brest Jun 2025
Canonical Metrics on Families of Vector Bundles

SEMINAR TALKS

Geometry, Analysis and Mathematical Physics Seminar, York November 2026
Semistability and asymptotics of geometric flows
Centre of Quantum Mathematics Research Seminar, Syddansk Universitet November 2026
Semistability and asymptotics of geometric flows
Purdue Geometry and Geometric Analysis Seminar, Purdue (online) October 2026
Semistability and asymptotics of geometric flows

JUNIOR SEMINAR TALKS

Junior Analysis and Probability Seminar, Warwick Jun 2026
Semistability and asymptotics of geometric flows
Hodge Club, Edinburgh Apr 2026
Groups acting on symplectic manifolds and moduli theory

Postgraduate Seminar, Warwick <i>Symmetries and dynamics - Introduction to symplectic geometry</i>	<i>Nov 2025</i>
Junior Algebraic Geometry Seminar, Warwick <i>Canonical metrics on families of vector bundles</i>	<i>May 2025</i>

OTHER TALKS

Reading group on <i>Non-abelian Hodge correspondence</i> <i>Non-abelian Hodge correspondence and hyperkähler geometry</i>	<i>Jun 2026</i>
Reading group on <i>K-stability of Fano varieties</i> <i>Θ-stratifications</i>	<i>Jun 2026</i>
Reading group on <i>Gauge theoretic invariants</i> <i>Compactness and wall-crossing</i>	<i>Nov 2025</i>
Reading group on <i>Moduli of weighted hyperplane arrangements</i> <i>Introduction to toric varieties</i>	<i>Nov 2025</i>
Reading group on <i>Gauge theoretic invariants</i> <i>Introduction to Gauge Theory and Seiberg–Witten invariants</i>	<i>Oct 2025</i>
Reading group on <i>K-stability of Fano Varieties</i> <i>The α- and δ-invariants</i>	<i>Jun 2025</i>
Glasgow Geometry and Topology Pre-Seminar <i>Gauge theory and Floer theory</i>	<i>Mar 2025</i>
Reading group on <i>Positivity in Algebraic Geometry</i> <i>Big line bundles, volume and Zariski decomposition</i>	<i>Feb 2025</i>
Reading group on <i>Algebraic Stacks</i> <i>Deformation theory and smoothness</i>	<i>Feb 2025</i>
Reading group on <i>Geometric Invariant Theory</i> <i>The Kempf–Ness theorem</i>	<i>Dec 2024</i>

TEACHING

MA3F1 Introduction to Topology Support Classes and Marking	<i>Term 1 2026-27</i>
MA453 Lie Algebras Support Classes	<i>Term 2 2025-26</i>
MA259 Multivariable Calculus Marking	<i>Term 2 2025-26</i>

EXPERIENCE

Summer research internship , University of Cambridge Summer project supervised by Alexei Kovalev, studying hyperkähler metrics on coadjoint orbits.	<i>Summer 2023</i>
Summer research internship , University of Cambridge Summer project supervised by Anders Hansen, studying (non-)computability of optimisation problems.	<i>Summer 2022</i>

PRIZES AND SCHOLARSHIPS

Clemmow/B Talbot Prize, K Roth Prize <i>Awarded for highest first in Part II Mathematics at Peterhouse</i>	<i>2023</i>
Senior Burkill Prize for Mathematics, Routh prize for Mathematics <i>Awarded for highest first in Part IB Mathematics at Peterhouse</i>	<i>2022</i>
Summer Research Bursary <i>Funding from the Faculty of Mathematics for summer research projects.</i>	<i>2022, 2023</i>
Scholarship in Mathematics <i>Scholar of Peterhouse, University of Cambridge</i>	<i>2021-24</i>
College prize <i>College prize for academic achievement</i>	<i>2021, 2022, 2023, 2024</i>

CONFERENCES ATTENDED

Trends in Differential and Complex Geometry , Oxford	<i>Sep 2026</i>
Degenerations in Complex Geometry , Institut Mittag-Leffler	<i>Jun-Jul 2026</i>
CALF , Nottingham	<i>Jun 2026</i>
Complex Geometry in Bordeaux , Bordeaux	<i>May 2026</i>
Living on the edge of the moduli space , Aarhus	<i>May 2026</i>
CALF , Imperial College London	<i>Mar 2026</i>
Brussels-London Geometry Seminar , Brussels	<i>Jan 2026</i>
Integrable Day 2025 , Loughborough	<i>Nov 2025</i>
COW , Warwick	<i>Nov 2025</i>
VII BrAG Meeting , Nottingham	<i>Sep 2025</i>
Singular canonical Kähler metrics on compact and non-compact manifolds , Budapest	<i>Sep 2025</i>
Summer Research Institute in Algebraic Geometry , Colorado	<i>Jul 2025</i>
Fibrations and Deformations , Brest	<i>Jun 2025</i>
Winter School on K-stability , CIRM, Luminy	<i>Mar 2025</i>
UKAG Network Winter School , Lancaster	<i>Dec 2024</i>
GLEN , Glasgow	<i>Dec 2024</i>

SKILLS

Programming Languages

Haskell, Python, Julia, Mathematica, Rust

Languages

Native: English, Cantonese, Mandarin

Basic: French, Spanish

Last updated: September 28, 2026